INCH-POUND

MIL-DTL-2156D
2 April 2010
SUPERSEDING
MIL-C-2156C (PA)
7 MAY 1976

DETAIL SPECIFICATION

CHARGE, DEMOLITION, SHAPED, 15 LB. - M2A4
LOADING, ASSEMBLING AND PACKING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the loading, assembling and packing for one type of demolition kit designated as Charge, Demolition, Shaped, 15 lb.– M2A4.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

Comments, suggestions, or questions on this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: RDAR-QES-E, Picatinny Arsenal, New Jersey 07806-5000 or emailed to ardecstdzn@conus.army.mil. Since contact information can change, you may want to verify the currency of this information using ASSIST Online database at https://assist.daps.dla.mil.

AMSC N/A FSC 1375
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DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1168	-	Ammunition Lot Numbering and Ammunition Data Card
MIL-STD-1916	-	DOD Preferred Method for Acceptance of Product
MIL-STD-2073-1	-	Standard Practice for Military Packaging

(Copies of these documents are available online at https://assist.daps.dla.mil/quicksearch/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094).

2.2.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation (see 6.2).

DRAWINGS

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

8861565	-	Charge, Demolition, Shaped, 15 lb., M2A4 Loading Assembly
8861569	-	Body Assembly
8858393	-	Booster
9281170	-	Packing and Marking for Charge, Demolition, Shaped, 15 lb., M2A4

(Copies of these drawings are available from U.S. Army ARDEC, ATTN: RDAR-EIS-PE, Picatinny, NJ 07806-5000, or email pica.drawing.request@conus.army.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM International

ASTM E 1742	-	Standard Practice for Radiographic Examination

(Copies of this document are available from http://www.astm.org or ASTM International, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

National Aerospace Standard

NAS 410	-	Certification and Qualification of Nondestructive Test Personnel

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(Copies of this document are available from http://www.aia-aerospace.org/ or Aerospace Industries Association, 1000 Wilson Boulevard, Suite 1700, Arlington, VA 22209-3928.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contractual document or purchase order (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.1.1 Radiographic examination. Radiographic examination in accordance with 4.3.3.3 shall be conducted on first article samples that are to be subjected to the Functioning Test, prior to the execution of the Functioning Test in accordance with Table II. Radiographic examination data shall not be used to accept/reject this system.

3.2 Conformance inspection. A sample of M2A4 shaped charges shall be subjected to conformance inspection in accordance with 4.3.

3.3 Material. Materials and parts shall comply with drawings cited in 4.3.2.1 – 4.3.2.9.

3.3.1 Specific gravity of booster. The specific gravity of the booster shall be minimum 1.50 when tested in accordance with 4.3.3.1.

3.3.2 Average weight of loading assemblies. The average weight of the loading assemblies shall be not less than 10.1 lbs when tested in accordance with 4.3.3.2 (Minimum 10 lbs of Comp B plus 48 grams of booster material).

3.4 Assemblies. The assemblies shall comply with all requirements specified on 8861565, and with all requirements specified in referenced specifications.

3.5 Functioning test. The loaded 15 lb Shaped Charge shall detonate high order when initiated by a M6 or M7 blasting cap or equivalent military initiation device.

3.6 Workmanship. All parts and assemblies shall be clean and free of explosive on exterior, chips, cracks, crazes, dirt, grease, rust, corrosion products, and other defects and foreign matter which would affect their serviceability. The cleaning method used shall not be injurious to any part or assembly nor shall the cleaning agent contaminate them. Exterior surface coatings shall be continuous except for a few light scratches not exposing the base material. All required markings shall be neat and sharply defined.

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4. VERIFICATION

Table I. Requirements/verification cross reference matrix

Method of Verification	Classes of Verification
1- Analysis 2- Examination 3- Test	A - First article B - Conformance

Section 3 Requirements	Description	Verification Method			Verification Class		Section 4
		1	2	3	A	B	
3.1	First article	X	X	X	X		4.2
3.1.1	Radiographic examination		X				4.3.3.3
3.2	Conformance inspection			X		X	4.3
3.3	Material	X	X	X	X	X	4.3.2.1-4.3.2.9
3.3.1	Specific gravity of booster	X			X	X	4.3.2.1, 4.3.3.1
3.3.2	Average weight of loading assemblies		X		X	X	4.3.2.3, 4.3.3.2
3.4	Assemblies		X	X	X	X	4.3
3.5	Functioning test			X	X	X	4.3.2.4, 4.3.3.4
3.6	Workmanship		X		X	X	4.3

4.1 Classification of verification. The verification requirements specified herein are classified as follows:

a. First Article Inspection (see 4.2)

b. Conformance Inspection (see 4.3)

4.2 First article.

4.2.1 Submission. When specified, a sample of the shaped charge and subcomponents shall be subjected to first article inspection in accordance with provisions of 4.2.2 and Table II.

4.2.2 Inspections to be performed. The first article samples shall be subjected to all of the examinations and tests specified in Table II.

4.2.2.1 Rejection. The first article sample shall be rejected if any of the inspections/tests fail to meet specification or drawing requirements.

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Table II. First article inspection

Examination or test	Conformance criteria	Requirement paragraph	Paragraph reference/ inspection method
Radiographic examination	10, see 1/	3.1.1	4.3.3.3
Functioning Test	10, see 1/	3.4	4.3.3.4
Parts Assemblies			
Examination of defects (where applicable) and Average Weight	25	3.3	4.3.2.2 & 4.3.2.3 4.3.3.2
Booster			
Examination of defects and Specific Gravity (8858393)	10	3.2	4.3.2.1 and 4.3.3.1
Loading Assemblies			
Examination of defects (where applicable) and Weight of Charge (8861565)	25	3.3	4.3.2.3 thru 4.3.2.5 4.3.3.2
Fiberboard and Packing boxes			
Examination of defects (where applicable) and heat seal test of sealed bag (9281170)	10	3.2	4.3.2.5 thru 4.3.2.9
NOTE: 1/ Radiographic examination shall be conducted on the First Article samples that are to be subjected to the Functioning Test, prior to the execution of the Functioning Test. Radiographic examination data will not be used to accept/reject this system. Any test unit which fails to detonate high order shall be classified as defective.			

4.3 Conformance inspection.

4.3.1 Inspection lot formation. Inspection lots shall comply with the lot formation provision of MIL-STD-1916. In addition, inspection lots of loaded shaped demolition charges shall contain:

a. Parts or part assemblies from not more than one lot interfix number from one manufacturer.

b. Composition B of the same grade from one lot interfix number from one manufacturer which contains TNT from not more than one lot interfix number from one manufacturer.

c. Composition A-3 from not more than two lots from one manufacturer.

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All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168.

4.3.2 Examination. Unless otherwise specified in the Conformance Inspection, sampling plans for major and minor defects shall be in accordance with MIL-STD-1916, Verification Level IV for Major defects and Verification Level II for Minor defects.

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4.3.2.1	Booster			Drawing number 8858393 Next higher assembly 8861565
Category	Examination or test	Conformance criteria	Requirement paragraph	Paragraph reference/ inspection method
Critical	None defined			
Major				
101	Concentricity of small cavity to outside diameter	IV	3.3	Gage
102	Diameter of small cavity, minimum (min.)	IV	3.3	Gage
103	Depth of Large cavity, min.	IV	3.3	Gage
104	Specific gravity	5	3.3	4.3.3.1
Minor				
201	Depth of small cavity, min.	II	3.3	Gage
202	Diameter of large cavity, min.	II	3.3	Gage
203	Total length, maximum (max.)	II	3.3	Gage
204	Radius missing	II	3.3	Visual
205	Evidence of poor workmanship	II	3.6	Visual

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4.3.2.2	Body assembly prior to loading			Drawing number 8861569 Next higher assembly 8861565
Category	Examination or test	Conformance criteria	Requirement paragraph	Paragraph reference/ inspection method
Critical	None Defined			
Major				
101	Outside diameter, max.	IV	3.4	Gage
102	Interior protective finish, discontinuous, scratched or contains bare spots	IV	3.4	Visual
103	Foreign matter present in interior	IV	3.4	Visual
104	Carrying strap damage	IV	3.4	Visual
Minor				
201	Evidence of poor workmanship	II	3.6	Visual

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4.3.2.3	HE loading assembly prior to assembly of closing cap			Drawing number 8861565 Next higher assembly 8861565
Category	Examination or test	Conformance criteria	Requirement paragraph	Paragraph reference/ inspection method
Critical	None Defined			
Major				
101	Component missing, loose or improperly assembled	IV	3.4	Visual/Manual
102	Body assembly damaged	IV	3.4	Visual
103	Average weight	25	3.4	4.3.3.2
Minor				
201	Depth to charge, min.	II	3.4	Gage
202	Evidence of poor workmanship	II	3.6	Visual

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4.3.2.4	Assembly (as applicable)			Drawing number 8861565 Next higher assembly
Category	Examination or test	Conformance criteria	Requirement paragraph	Paragraph reference/ inspection method
Critical	None defined			
Major				
101	Gage does not enter detonator well freely to full length	IV	3.4	Visual
102	Closing cap assembly loose	IV	3.4	Manual
103	Protective tape missing	IV	3.4	Visual
104	Functioning Test	10, see 1/	3.5	4.3.3.4
Minor				
201	Total length	II	3.4	Gage
202	Marking missing, incorrect or illegible	II	3.4	Visual
203	Color coding improper (see Dwg. 8861569)	II	3.4	Visual
204	Evidence of poor workmanship	II	3.6	Visual

NOTE:
1/ Any test unit which fails to detonate high order shall be classified as defective. After three (3) consecutive lots have met the criteria, a sample of five (5) test units shall be selected and function tested out of each lot produced. If one or more loaded units fail to function, the sampling shall return to the original test plan and continue until three (3) consecutive lots have complied with the acceptance criteria specified.

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4.3.2.5	Box, fiberboard prior to sealing			Drawing number 9281170 Next higher assembly
Category	Examination or test	Conformance criteria	Requirement paragraph	Paragraph reference/ inspection method
Critical	None defined			
Major				
101	Component missing or improperly packed	IV	3.3	Visual
Minor				
201	Evidence of poor workmanship	II	3.6	Visual

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4.3.2.6	Box, fiberboard			Drawing number 9281170 Next higher assembly
Category	Examination or test	Conformance criteria	Requirement paragraph	Paragraph reference/ inspection method
Critical	None defined			
Major				
101	Sealing strip incomplete or loose	IV	3.3	Visual
102	Box punctured, torn or cut	IV	3.3	Visual
103	Corner not blunted	IV	3.3	Visual
Minor				
201	Marking missing, incorrect or illegible	II	3.3	Visual
202	Contents loose	II	3.3	Manual
203	Evidence of poor workmanship	II	3.6	Visual

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4.3.2.7	Sealed packed barrier bag prior to packing in box			Drawing number 9281170 Next higher assembly
Category	Examination or test	Conformance criteria	Requirement paragraph	Paragraph reference/ inspection method
Critical	None defined			
Major				
101	Waterproofness of barrier bag destroyed by rupture, puncture or separation of heat seal	IV	3.3	Visual
102	Evidence of crushed box (see 1/)	IV	3.3	Visual
103	All excess air not exhausted from bag prior to sealing	IV	3.3	Visual
104	Heat seal test (4.3.3.5)	IV	3.3	Gage
Minor				
201	Heat seal of barrier bag located so that one or more subsequent seals cannot be applied	II	3.3	Visual
202	Marking of bag improper or unidentifiable	II	3.3	Visual
203	Evidence of poor workmanship	II	3.6	Visual
Note: 1/ Package should be opened if there is evidence of crushing.				

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4.3.2.8	Packing box prior to securing top			Drawing number 9281170 Next higher assembly
Category	Examination or test	Conformance criteria	Requirement paragraph	Paragraph reference/ inspection method
Critical	None defined			
Major				
101	Number of sealed bags incorrect	IV	3.3	Visual
102	Any packing component missing	IV	3.3	Visual
Minor				
201	Insufficient filler material	II	3.3	Visual
202	Evidence of poor workmanship	II	3.6	Visual

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4.3.2.9	Packing box			Drawing number 9281170 Next higher assembly
Category	Examination or test	Conformance criteria	Requirement paragraph	Paragraph reference/ inspection method
Critical	None defined			
Major				
101	Board broken or split	IV	3.3	Visual
102	Contents exposed or loose	IV	3.3	Visual/Manual
103	Hardware or strapping missing, loose or broken	IV	3.3	Visual/Manual
Minor				
201	Marking missing, incorrect or illegible	II	3.3	Visual
202	Evidence of poor workmanship	II	3.6	Visual

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4.3.3 Test Methods and Procedures.

4.3.3.1 Specific gravity of booster. (See Dwg. 8858393) A sample of five (5) boosters shall be selected from each lot for this test. If any booster fails to meet the requirement specified, the lot shall be rejected. The booster shall be weighed on a balance and the weight recorded as dry weight. The booster shall then be weighed while completely submerged in water, taking care to remove all air bubbles remaining on the surfaces of the booster. The temperature of the water shall be 68 degrees Fahrenheit (^{o}F) +/- 2 ^{o}F. This weight shall be recorded as weight under water. (When submerged, the position of the booster shall be such that no air is entrapped.) Specific gravity shall be calculated as follows:

$$\textit{Specific gravity} = \frac{\textit{Weight of booster dry}}{\textit{(Weight of booster dry minus Weight of booster immersed in water)}} \times \textit{Specific gravity of } H_2O \textit{ at test temperature (see Table III)}$$

Table III. Specific gravity of water in test temperature range.

Temp °F	Temp °C	Specific Gravity of H_2O
66	18.89	0.9984
67	19.44	0.9983
68	20.00	0.9982
69	20.56	0.9981
70	21.11	0.9980

4.3.3.2 Weight of charge and average weight. The loading assembly shall be weighed on a scale capable of determining weights within an accuracy of two (2) ounces. The average weight shall be obtained from twenty-five (25) parts assemblies (dwg. 8861569) and subtracted from the weight of the loading assembly (dwg. 8861565) to determine compliance with the requirement. Any assembly failing to meet the weight requirement shall be classed defective and the lot shall be rejected.

4.3.3.3 Radiographic examination. Radiographic examination shall be conducted on the first article samples (see 4.2.2.2) that are to be subjected to the Functioning Test prior to the execution of the Functioning Test. Ten (10) test units shall be selected from each lot. The first article sample shall be subjected to radiographic examination for detection of voids, cracks, and foreign material. Examinations of the radiograph shall be made to determine excessive cavitation, separation, and cracking of the explosive and presence of any foreign material. Radiographic equipment, operations and procedures shall be qualified in accordance with ASTM E 1742. Radiographic personnel shall be qualified in accordance with NAS 410.

4.3.3.4 Functioning test. Ten (10) test units shall be selected from each lot. The test shall be detonated using a M6 or M7 blasting cap or equivalent military initiation device. Any test unit which fails to detonate high order shall be classified as defective. After three (3) consecutive lots

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have met the criteria, a sample of five (5) test units shall be selected and function tested out of each lot produced. If one or more loaded units fail to function, the sampling shall return to the original test plan and continue until three (3) consecutive lots have complied with the acceptance criteria specified.

4.3.3.5 Heat seal test of sealed bag (destructive). Sampling, acceptance criteria and test procedures for this test shall be as specified in MIL-STD-2073-1, Method 40.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DOD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The 15 lb. Shaped Demolition Charge is primarily employed for burring holes in earth, metal, concrete, and paved or unpaved roads

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Provisions for submission of first article samples.

c. Requirements for Ammunition Lot Numbering, if applicable.

d. Requirement for acceptance inspection equipment (AIE) designs.

e. Certificate of Conformance for each lot or shipment of product.

f. Preservation and packaging. Preservation, packaging, packing and marking should be in accordance with Dwg. 9281170.

6.3 Submission of inspection equipment designs for approval. Equipment designs should be submitted per the Contract Data Requirements Lists (DD Form 1423).

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6.4 Radiographic examination. Complete results of the radiographic examination should be available to the Government through the Contracting Officer. Distribution should include: U.S. Army RDECOM-ARDEC, RDAR-MEM-G, Picatinny, NJ 07806-5000.

6.5 Certification. Unless otherwise specified by the Contracting Officer, certification should be submitted by the contractor in accordance with the contract. Distribution should include: U.S. Army ARDEC, RDAR-MEM-G, Picatinny, NJ 07806-5000.

6.6 Subject term (key word) listing.

Explosive
Composition B

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodian:
Army-AR
Navy -OS
Air Force - 70

Preparing Activity:
Army-AR
(Project 1375-2010-006)

Review Activity
Navy – AS, NP
Air Force – 11, 99

Civil agency:
GSA-FAS

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